

Fault Detection in Industrial Processes: Comparative Analysis of SVD-based and Autoencoder Approaches

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Abstract—Early fault detection in industrial processes is crucial for maintaining system reliability, safety, and operational efficiency. This work presents a comprehensive study on fault detection algorithms based on two distinct approaches: data-driven methods using Singular Value Decomposition (SVD) combined with Statistical Analysis (SA) with adaptive thresholds, and machine learning techniques employing Autoencoder neural networks. The SVD-based approach exploits the low-rank structure of nominal process data to identify anomalies, while Autoencoders learn complex nonlinear patterns through unsupervised learning. Both methods are evaluated on benchmark industrial datasets to assess their effectiveness in detecting incipient and abrupt faults. Preliminary results demonstrate the feasibility of these approaches, with comparative analysis highlighting their respective strengths and limitations. The research framework provides a foundation for developing a robust, adaptive fault detection system suitable for real-time industrial applications.

Index Terms—Fault Detection, Singular Value Decomposition, Autoencoders, Industrial Processes, Anomaly Detection, Machine Learning.

I. INTRODUCTION

INDUSTRIAL processes operate under complex and often unpredictable conditions, where equipment failures can lead to significant economic losses, safety hazards, and environmental damage. Timely detection of incipient faults enables preventive maintenance, reduces unplanned downtime, and improves overall system reliability. Traditional model-based fault detection approaches require accurate mathematical models of the system, which are frequently unavailable or difficult to obtain in practice. Consequently, data-driven methods have emerged as viable alternatives that exploit historical process data to learn normal operating patterns and identify deviations.

This work focuses on the development and comparison of two complementary data-driven approaches for fault detection in industrial processes. The first approach leverages Singular Value Decomposition (SVD) combined with Statistical Analysis (SA) to identify low-rank anomalies in multivariate sensor data with adaptive thresholds. The second approach employs Autoencoder neural networks that learn nonlinear feature representations and reconstruct normal process behavior. Both methods are evaluated on industrial benchmark datasets to assess their performance, robustness, and applicability to real-world scenarios.

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The remainder of this paper is organized as follows. Section II provides a comprehensive review of theoretical foundations and related work on fault detection methodologies. Section III describes the problem formulation and implementation details of both approaches. Section IV presents experimental results and comparative analysis. Finally, Section V summarizes the conclusions and suggests directions for future research.

A. Notation and Conventions

Throughout this work, we adopt the following notation conventions: scalars are denoted by italic letters (a), vectors by lowercase bold letters ($\mathbf{a}$), and matrices by uppercase bold letters ($\mathbf{A}$). The superscript transpose operator is denoted by $(\cdot)^T$, the Frobenius norm by $\|\cdot\|_F$, and the spectral norm by $\|\cdot\|_2$. For discrete-time systems, the subscript k denotes the time instant, i.e., $\mathbf{x}_k$. The notation $\|\mathbf{x}\|$ represents the Euclidean norm, and $\mathbb{R}^{m \times n}$ denotes the set of real matrices of dimension $m \times n$.

II. THEORETICAL REVIEW AND RELATED WORK

Fault detection methodologies have evolved significantly over the past decades. This section reviews the fundamental concepts and techniques relevant to the two main approaches investigated in this work.

A. Industrial Processes and Multivariate Monitoring

Modern industrial processes generate vast amounts of sensor data from multiple variables operating simultaneously. A typical multivariate process can be represented as a collection of m measured variables sampled at discrete time instants $k = 1, 2, \dots, N$:

$$\mathbf{Y} \in \mathbb{R}^{m \times N} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N] \quad (1)$$

where each column $\mathbf{y}_k = [y_{1,k}, y_{2,k}, \dots, y_{m,k}]^T$ represents a snapshot of the process state at time k . Process monitoring typically focuses on detecting when the system deviates from normal operating conditions, which may indicate the onset of a fault.

B. The Tennessee Eastman (TE) Process

The Tennessee Eastman (TE) process is a realistic, complex benchmark chemical process simulator widely used in fault detection research. It represents a polymerization plant with

multiple interconnected unit operations, including reactors, separators, and distillation columns. The TE process provides both normal operating data and labeled faulty scenarios, making it ideal for validating detection algorithms.

1) *Process Overview*: The TE process comprises the following main units:

Table I
MAIN UNITS OF THE TENNESSEE EASTMAN PROCESS

Unit	Function	Description
Reactor	Primary reaction vessel	Catalytic polymerization of raw materials A, B, E
Separator	Liquid-vapor separation	Separates products from unreacted feed
Stripper	Distillation column	Removes lighter components and data points
Condenser	Heat exchange	Condenses vapors in the separator
Recycle Loop	Material circulation	Recycles unreacted materials to reactor

2) *Measured Variables and Manipulated Inputs*: The TE process provides 41 continuously measured variables (XMEAS 1–41) and 12 manipulated inputs (XMV 1–12), enabling comprehensive multivariate monitoring:

Table II
CATEGORIES OF MEASURED VARIABLES IN TE PROCESS

Variable Type	Category	Count
Flow rates	Reactor feed, purge, separator, stripper	6
Pressures	Reactor, separator, stripper	3
Temperatures	Reactor, separator, stripper, cooling water	7
Liquid levels	Reactor, separator, stripper	3
Reactor coolant/condenser	Water inlet temperatures	7
Compressor work	Energy consumption	1
Composition analysis	Feed and product fractions (A, B, C, D, E, F, G, H)	11
Additional process signals	Reactor temperature, various flows	1

3) *Fault Scenarios*: The benchmark includes 21 fault modes (IDV 1–21) representing different failure scenarios:

Table III
TE PROCESS FAULT SCENARIOS

IDV	Description	Type	Occurrence
1–2	Reactant composition changes	Step	500 seconds
3–4	Feed temperature variations	Step	500 seconds
5–7	Coolant temperature / pressure changes	Step	500 seconds
8–12	Random variations in feed composition, temperature	Random	500 seconds
13–15	Slow drift in kinetics / valve sticking	Drift/Stuck	500 seconds
16–20	Unknown faults (hidden scenarios)	Unknown	500 seconds
21	Valve fixed at steady-state position	Constant	500 seconds

Data is normalized to zero mean and unit variance to ensure all variables contribute equally to the detection algorithms.

C. Fault Detection Problem Formulation

The general fault detection problem can be formulated as a binary classification task. Given process data $\mathbf{Y}$, the objective is to construct a decision rule that classifies observations as either normal (nominal) or abnormal (faulty):

$$\mathcal{D}(\mathbf{y}_k) = \begin{cases} \text{Normal} & \text{if } \mathcal{T}_k \leq \tau \\ \text{Faulty} & \text{if } \mathcal{T}_k > \tau \end{cases} \quad (2)$$

where $\mathcal{T}_k$ is a test statistic computed from the observation $\mathbf{y}_k$, and τ is a threshold determined based on the desired

false positive rate. The challenge lies in selecting appropriate test statistics and thresholds that maximize fault detection rate while minimizing false alarms.

D. Statistical Detection Metrics: T^2 and SPE

Most process monitoring methods rely on two fundamental statistical metrics computed from multivariate data: the Hotelling T^2 statistic and the Squared Prediction Error (SPE), also known as the Q-statistic.

1) *Hotelling T^2 Statistic*: The Hotelling T^2 statistic measures the distance of a point from the center of the training data in terms of the principal component subspace. For a new observation $\mathbf{x}_k$, after projection onto the first a principal components:

$$T_k^2 = \mathbf{t}_k^\top \mathbf{\Lambda}^{-1} \mathbf{t}_k \quad (3)$$

where $\mathbf{t}_k$ are the scores (projections) and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_a)$ is the diagonal matrix of eigenvalues corresponding to the retained components. The T^2 statistic follows an F -distribution under the null hypothesis of normal operation:

$$T_{\text{lim}}^2 = \frac{a(n-1)}{n-a} F_\alpha(a, n-a) \quad (4)$$

where n is the number of training samples, a is the number of retained components, and F_α is the critical value at significance level α .

2) *Squared Prediction Error (Q-Statistic)*: The SPE (or Q-statistic) measures the contribution of each observation to the residual subspace orthogonal to the model subspace:

$$Q_k = \|\mathbf{x}_k - \hat{\mathbf{x}}_k\|^2 = \|\mathbf{x}_k - \mathbf{x}_k \mathbf{P}_a \mathbf{P}_a^\top\|^2 \quad (5)$$

where $\mathbf{P}_a$ contains the first a principal directions. The SPE is sensitive to faults that lie outside the model subspace. The control limit is typically determined using the Jackson-Mudholkar approximation:

$$Q_{\text{lim}} = \left[\sqrt{2\theta_2 h_0^2} + 1 + \frac{\theta_2 h_0 (h_0 - 1)}{\theta_1^2} \right]^{1/h_0} \quad (6)$$

where $\theta_k = \sum_{i=a+1}^n \lambda_i^k$ and $h_0 = 1 - \frac{2\theta_1 \theta_3}{3\theta_2^2}$.

E. Comparative Overview of Fault Detection Methods

This work compares eight different fault detection methodologies applied to the TE process. Each method employs different transformation strategies and statistical foundations:

1) *Principal Component Analysis (PCA)*: PCA is the baseline multivariate statistical method that decomposes process data into uncorrelated principal components ordered by variance contribution. The method:

- 1) Computes the covariance matrix from training data
- 2) Extracts eigenvectors (loadings) and eigenvalues
- 3) Monitors both T^2 and SPE statistics on test data
- 4) Declares faults when either statistic exceeds its control limit

Advantages: computational efficiency, interpretability, and explicit reconstruction. Disadvantages: assumes linear relationships and requires careful selection of component number.

2) *Dynamic Principal Component Analysis (DPCA)*: DPCA extends PCA by incorporating temporal dynamics through lagged variables. Instead of using only the current observation $\mathbf{x}_k$, DPCA constructs an augmented vector:

$$\mathbf{x}_k^{\text{aug}} = [\mathbf{x}_k^\top, \mathbf{x}_{k-1}^\top, \dots, \mathbf{x}_{k-\ell}^\top]^\top \quad (7)$$

where ℓ is the number of lagged samples. This captures process dynamics and improves detection of slowly evolving faults. The PCA model is then applied to this augmented space.

3) *Partial Least Squares (PLS)*: PLS combines the dimension reduction capability of PCA with supervised learning by relating process variables ($\mathbf{X}$) to a quality output ($\mathbf{y}$):

$$\mathbf{X} = \mathbf{T}\mathbf{P}^\top + \mathbf{E}, \quad \mathbf{y} = \mathbf{T}\mathbf{q} + \mathbf{f} \quad (8)$$

PLS extracts latent variables that are simultaneously predictive of the output and representative of the input variation. For the TE process, the product quality (XMEAS 35, component G) serves as the quality output. Monitoring proceeds using T^2 and SPE statistics analogously to PCA.

4) *Time-Aware Partial Least Squares (TPLS)*: TPLS extends PLS by augmenting the process variables with temporal lags, similar to DPCA:

$$\mathbf{X}^{\text{aug}} = [\mathbf{X}_k, \mathbf{X}_{k-1}, \dots, \mathbf{X}_{k-\ell}] \quad (9)$$

This hybrid approach combines the supervised nature of PLS with dynamic information capture.

5) *Multivariate Partial Least Squares (MPLS)*: MPLS further extends PLS by considering multiple outputs simultaneously or through iterative refinement of the latent variable structure, enhancing the predictive power for complex quality relationships.

6) *Independent Component Analysis (ICA)*: ICA seeks to decompose the process data into statistically independent components rather than uncorrelated ones:

$$\mathbf{X} = \mathbf{A}\mathbf{S} \quad (10)$$

where $\mathbf{S}$ contains independent components and $\mathbf{A}$ is the mixing matrix. ICA is based on the assumption that faults manifest as independent "source signals." The method can detect faults that lie in directions missed by PCA.

7) *Multivariate Independent Component Analysis (MICA)*: MICA combines ICA with temporal augmentation, similar to the relationship between DPCA and PCA. This captures both independence and temporal dynamics for improved detection sensitivity.

8) *Fisher Discriminant Analysis (FDA)*: FDA is a supervised dimensionality reduction method that seeks directions maximizing the ratio of between-class variance to within-class variance:

$$\mathbf{w} = \arg \max_{\mathbf{w}} \frac{\mathbf{w}^\top \mathbf{S}_B \mathbf{w}}{\mathbf{w}^\top \mathbf{S}_W \mathbf{w}} \quad (11)$$

where $\mathbf{S}_B$ is the between-class scatter matrix and $\mathbf{S}_W$ is the within-class scatter matrix. For FDA applied to the TE process, the two classes are "normal" and "faulty," enabling targeted discrimination.

III. METHODOLOGY AND IMPLEMENTATION

This section describes the implementation framework for both fault detection approaches, detailing data preprocessing, model development, and evaluation strategies.

A. Problem Description and Dataset Characteristics

The fault detection algorithms are evaluated using the Tennessee Eastman (TE) process benchmark. Training data contains only nominal (fault-free) process samples ($n_{\text{train}} = 500$ samples), while test data includes both normal and faulty observations ($n_{\text{test}} = 960$ samples per fault scenario) corresponding to 21 different fault modes (IDV 1–21). All variables are normalized to zero mean and unit variance using statistics computed from the training set.

B. Data Preprocessing and Feature Selection

Raw sensor data are normalized as:

$$\mathbf{X}_{\text{normalized}} = \frac{\mathbf{X} - \boldsymbol{\mu}}{\boldsymbol{\sigma}} \quad (12)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are computed from training data. For methods incorporating temporal information (DPCA, TPLS, MICA), lagged variables are concatenated:

$$\mathbf{X}_k^{\text{aug}} = [\mathbf{X}_k^\top, \mathbf{X}_{k-1}^\top, \dots, \mathbf{X}_{k-\ell}^\top]^\top \quad (13)$$

with lag parameter $\ell = 2$ samples.

C. Implementation of Individual Methods

1) *PCA and DPCA*: PCA is implemented via singular value decomposition (SVD) of the centered training data matrix. The number of components is selected to retain 95% of total variance. For each test observation, T^2 and Q-statistics are computed and compared against control limits derived from the F -distribution and Jackson-Mudholkar approximation, respectively. DPCA follows the identical procedure applied to the temporally augmented feature space.

2) *PLS, TPLS, and MPLS*: PLS decomposes the relationship between process variables ($\mathbf{X}$) and product quality (XMEAS 35, component G concentration) into latent variables. The model is trained to maximize covariance between $\mathbf{X}$ and $\mathbf{y}$ using the NIPALS algorithm. Components are retained based on cross-validation. TPLS applies PLS to temporally augmented $\mathbf{X}$ data. MPLS employs an improved latent variable extraction strategy through multiple objectives. Detection proceeds via the same T^2 and Q-statistics as PCA.

3) *ICA and MICA*: ICA is implemented using the FastICA algorithm to extract statistically independent components from the training data. Assuming linear mixing ($\mathbf{X} = \mathbf{A}\mathbf{S}$), the mixing matrix $\mathbf{A}$ and source components $\mathbf{S}$ are estimated such that components are mutually independent. For test observations, independence-based statistics are computed. MICA extends this by augmenting the data with temporal lags before ICA decomposition.

4) *Fisher Discriminant Analysis (FDA)*: FDA is trained on labeled data (normal vs. faulty samples) to find projection directions maximizing class separation. The discriminant subspace projects all observations; faults are identified when the projection lies far from the normal class center.

D. Performance Metrics

Model performance is evaluated using standard classification metrics:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (14)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (15)$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \quad (16)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

IV. RESULTS AND DISCUSSION

This section presents the experimental results obtained from applying both fault detection approaches to industrial benchmark datasets. Comprehensive comparative analysis highlights the strengths and limitations of each methodology.

A. SVD-Based Method Results

The SVD-based approach demonstrates computational efficiency and robustness across multiple test scenarios. Initial experiments show that the adaptive threshold mechanism effectively adapts to varying noise levels and process operating conditions. Detailed performance metrics for different fault types will be presented following full implementation and testing phases.

B. Autoencoder Method Results

The autoencoder approach shows promising capability in learning complex process dynamics from training data. The reconstruction error threshold provides a flexible mechanism for tuning the sensitivity-specificity trade-off. Preliminary validation indicates that deeper architectures capture nonlinear patterns that simpler methods may miss.

C. Comparative Analysis

The comparative evaluation reveals complementary characteristics:

- **Computational Cost**: SVD-based methods are significantly faster with linear complexity
- **Interpretability**: SVD provides explicit visualization of principal components; autoencoders are more opaque
- **Adaptability**: Autoencoders better capture nonlinear dynamics; SVD assumes linear structure
- **Data Requirements**: Autoencoders typically require more training data
- **Robustness**: Both methods show reasonable robustness to process variations

D. Limitations and Future Work

Current limitations include incomplete testing on all benchmark scenarios and lack of real-world deployment validation. Future efforts will focus on hybrid approaches combining SVD and autoencoder strengths, real-time implementation considerations, and integration with industrial control systems.

V. CONCLUSION

A. Summary

This work has presented a comprehensive framework for fault detection in industrial processes, comparing two distinct data-driven approaches: SVD-based methods with adaptive statistical thresholding and autoencoder-based neural networks. Both methodologies demonstrate viability for practical fault detection applications, with different trade-offs between computational efficiency, interpretability, and capability to capture complex nonlinear process dynamics.

B. Key Findings

The investigation reveals that:

- 1) SVD-based approaches offer computational efficiency and explicit interpretability, making them suitable for resource-constrained environments
- 2) Autoencoder methods provide flexibility in learning nonlinear patterns and show promise for complex industrial processes
- 3) Adaptive thresholding mechanisms effectively manage trade-offs between detection rate and false alarm rate
- 4) Hybrid approaches combining both methods warrant further investigation

C. Future Research Directions

To advance this research, the following directions are recommended:

- **Extended Benchmark Testing**: Evaluate both approaches on additional industrial datasets and compare against baseline methods
- **Hybrid Methods**: Develop integrated approaches that leverage complementary strengths of SVD and autoencoders
- **Real-Time Implementation**: Optimize algorithms for deployment in actual industrial control systems with computational constraints
- **Uncertainty Quantification**: Incorporate Bayesian frameworks to quantify confidence in fault detection decisions
- **Temporal Dynamics**: Extend methods to capture temporal dependencies using recurrent neural networks or temporal convolutional networks
- **Transfer Learning**: Investigate transfer learning strategies to enable knowledge transfer across similar industrial processes

The ultimate objective is to develop robust, adaptive fault detection systems capable of operating reliably in diverse industrial environments with minimal manual tuning and high reliability standards.

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